

The Program in Action

Figure 39-1 shows what the output will look like when you run *langtonsant.py*.

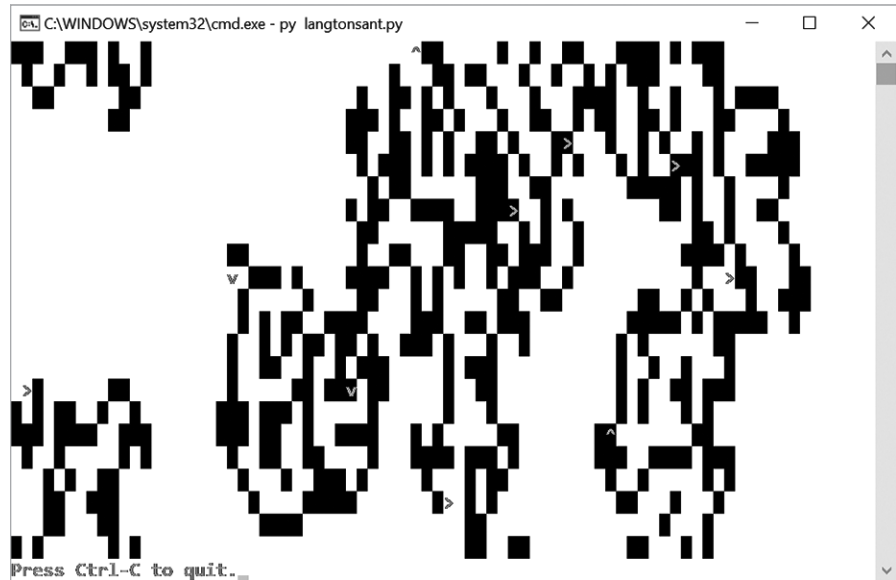


Figure 39-1: The hypnotic output of the Langton's Ant cellular automata

How It Works

This program uses two senses of “direction.” On the one hand, the dictionaries that represent each ant store *cardinal directions*: north, south, east, and west. However, turning left or right (or counterclockwise and clockwise, since we are viewing the ants from above) is a *rotational direction*. Ants are supposed to turn left or right in response to the tile they’re standing on, so lines 78 to 100 set a new cardinal direction based on the ant’s current cardinal direction and the direction they are turning.

```
1. """Langton's Ant, by Al Sweigart al@inventwithpython.com
2. A cellular automata animation. Press Ctrl-C to stop.
3. More info: https://en.wikipedia.org/wiki/Langton%27s_ant
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, artistic, bext, simulation"""
6.
7. import copy, random, sys, time
8.
9. try:
10.     import bext
11. except ImportError:
12.     print('This program requires the bext module, which you')
13.     print('can install by following the instructions at')
14.     print('https://pypi.org/project/Bext/')
15.     sys.exit()
16.
```